

Genetic Testing in Paediatric Liver Disease.

In the last few chapters we looked at ways to measure the liver: blood tests, imaging, stiffness and biopsy. This chapter is about a different kind of test. Instead of measuring the liver, we read the child's DNA and ask whether a single faulty gene explains the disease.

In paediatric hepatology this question comes up all the time. Many of the diseases we see in infants are inherited. Alagille syndrome, the progressive familial intrahepatic cholestasis diseases, alpha one antitrypsin deficiency, citrin deficiency, Wilson disease and the mitochondrial liver diseases are all caused by changes in single genes. Twenty years ago we had to test for them one gene at a time. Today a single sample of blood can be used to read dozens of genes, all the coding parts of the genome, or even the whole genome.

That sounds simple, but it is not. Genetic tests produce results that are easy to misread. A negative result does not always mean that there is no genetic disease. A positive result does not always mean that the variant causes the disease. And many results land in a grey zone in between, called a variant of uncertain significance.

The main idea of this chapter is this. A genetic test answers only the question it was asked, with the method it used, about the child it was given. To use genetic testing well, you need to know three things: what the test can read, who was tested, and how to interpret a result that is not clear.

There is a second idea that runs through the whole chapter. Never let a pending genetic result delay a treatment you can start today. Several inherited liver diseases have a fast biochemical clue and a treatment that works. In those diseases you treat first and let the gene confirm it later.

So this chapter asks five questions. First, what does each kind of genetic test read, and what does it miss? Second, how often does testing find a diagnosis, and why do the numbers vary so much? Third, which test should you send, and when should you send the parents too? Fourth, how do you handle a variant of uncertain significance, and when should old data be looked at again? And fifth, when does the gene actually change what you do for the child?

A short note on the sources. The most important document is the clinical practice guideline of the European liver society, EASL, on genetic cholestatic liver disease, published in the Journal of Hepatology in 2024. For the infant with jaundice we use the joint North American and European paediatric guideline, from NASPGHAN and ESPGHAN, written by Fawaz and colleagues and published in 2017. For how variants are classified we use the standards of the American College of Medical Genetics and

Genomics, called ACMG, from 2015, together with its statements on reanalysis from 2019 and on exome and genome sequencing from 2021. In the United Kingdom, the national genomic test directory of NHS England sets out who can have which test. Most of the numbers in this chapter come from single studies, and we will name them as we go.

A note on words. If you have not worked with genetics since medical school, these terms are worth a minute.

A gene is a stretch of DNA that carries the instructions for one protein. Most genes are split into pieces. The pieces that code for the protein are called exons. The pieces in between are called introns. All the exons together make up only about one to two percent of the genome.

A variant is any difference between the child's DNA and the reference sequence. Most variants are harmless. The word mutation is older, and it is now avoided because it sounds as if the change must be harmful.

A single-nucleotide variant is a change in one letter of the DNA. A copy-number variant is a larger change, in which a whole piece of DNA, sometimes a whole gene, is deleted or duplicated. You will often see deletions and duplications shortened to del dup.

A dominant disease needs only one faulty copy of the gene. Alagille syndrome is an example. A recessive disease needs both copies to be faulty. The progressive familial intrahepatic cholestasis diseases and Wilson disease are examples. When both copies are faulty we say the variants are biallelic.

A de novo variant is a new variant that is present in the child but in neither parent.

Phase means which copy of the gene a variant sits on. If a child has two variants in a recessive gene, they cause disease only if one sits on the copy from the mother and the other on the copy from the father. That is called in trans. If both sit on the same copy, the child is only a carrier.

Finally, the diagnostic yield is the proportion of children tested in whom the test finds a molecular diagnosis.

Part one. What each test reads.

There are five kinds of genetic test that a paediatric hepatologist will order. The easiest way to understand them is to ask, for each one, what it reads, what it misses, and how long it takes.

The first is single-gene testing, usually by Sanger sequencing. This reads one gene, letter by letter, very accurately. It misses every other gene, and it usually misses large deletions. The result comes back in one to three weeks. Today its use is narrow. It is the

right test when a variant is already known in the family and you want to know whether a sibling carries it. It is also useful for one gene in a clear clinical picture, for example ABCB4 in a child with small-duct sclerosing cholangitis. The EASL guideline of 2024 discourages using Sanger sequencing gene by gene as a first test for cholestasis.

The second is the cholestasis panel. A panel reads a fixed list of genes, usually forty to eighty, that are known to cause liver disease. Many panels also look for deletions and duplications. A panel misses any gene that is not on its list, including genes that were discovered after the panel was designed. It also misses variants deep inside introns. In a large American laboratory study, the median turnaround was twenty-one days. A panel is a good first test for an infant with a classic cholestatic picture.

The third is the exome. Exome sequencing reads all the exons of all the genes, the one to two percent of the genome that codes for protein. It misses most copy-number variants, the non-coding parts of the genome, repeated sequences and mitochondrial DNA. It takes several weeks. It works best as a trio, which means sequencing the child and both parents together. The exome is the right test for unexplained or multisystem disease. It has one great advantage over a panel. The data are stored, so they can be read again later when new genes are discovered.

The fourth is the genome. Genome sequencing reads the whole genome, coding and non-coding, and it picks up copy-number variants and mitochondrial DNA as well. What it still misses are changes in DNA methylation and some repeat expansions. It also produces more variants of uncertain significance to sort through. When it is done as a rapid trio genome, a preliminary result can come back in three to eight days. That makes it the test for a sick child in intensive care, for acute liver failure, and for any child where a transplant decision is waiting.

The fifth is the chromosomal microarray. This looks only for copy-number changes, the gains and losses of larger pieces of DNA. It cannot see single-letter changes. It takes one to three weeks. It is useful in a child with several congenital anomalies, where its yield is about ten percent.

Part two. Diagnostic yield, and why the numbers vary so much.

If you read the literature, you will find yields for genetic testing in paediatric liver disease that range from four percent to more than seventy percent. That is not because some tests are good and others bad. It is mainly because of who was tested. In other words, the yield measures the denominator.

Let us go through the studies, from the highest yield to the lowest.

In Japan, Togawa and colleagues, in 2016, tested thirty-one children whose disease already looked like a genetic cholestasis, using a panel. The yield was seventy-one

percent. In China, Chen and colleagues, in 2025, used exome sequencing in eighty children with unexplained liver disease, and found a diagnosis in fifty-seven point five percent. Fang and colleagues, in 2021, tested one hundred and seventy-two Chinese children with suspected inherited liver disease, using exome or panel, and the yield was fifty-five point eight percent. In Europe, Lenz and colleagues, in 2024, used exome sequencing in two hundred and sixty children with acute liver failure of unknown cause, and found a genetic diagnosis in thirty-seven percent. In Türkiye, Gürcan Kaya and colleagues, in 2025, tested three hundred and seventy-eight infants with neonatal cholestasis, and the yield was twenty-eight percent. In Japan, Ito and colleagues, in 2022, used a sixty-one-gene panel in one hundred and twenty-four infants with cholestasis, and the yield was twenty-six point six percent.

Now look at the large American laboratory studies, where any doctor could send a sample. Karpen and colleagues, in 2021, reported two thousand one hundred and seventy-one cholestasis panels, and the yield was twelve percent. Hoskins and colleagues, in 2026, reported ten thousand eight hundred and ninety-four panels from the same kind of service, and the definitive yield fell to six point eight percent. And in the Japanese study by Togawa, a group of forty-six children whose cholestasis came with prematurity and other non-genetic causes had a yield of only four point three percent.

So the yield is high when the children were carefully selected, and low when testing was opened to everyone. As the American panel was sent for more and more children, the yield fell from twelve percent to six point eight percent.

What about the type of test? In the study by Fang and colleagues, the panel found a diagnosis in sixty-two percent and the exome in forty-seven point two percent. That difference was not statistically significant. The selection of children mattered more than the platform.

It is also worth knowing which genes are found, because that depends on where you practise.

In the American laboratory studies, the most common genetic diagnoses were in JAG one and NOTCH two, which cause Alagille syndrome, ABCB eleven, SERPINA one, ABCB four and POLG. ABCC two and CFTR were also among the definite results. The lesson here is that many of these so-called diagnoses are mild conditions or co-factors, such as being a carrier for alpha one antitrypsin deficiency, or having Dubin–Johnson syndrome.

In the Japanese study of one hundred and twenty-four infants, three conditions made up seventy-eight point eight percent of all the genetic cholestasis: Alagille syndrome, neonatal Dubin–Johnson syndrome caused by ABCC two, and citrin deficiency caused by SLC twenty-five A thirteen. So any panel used in East Asia must include those genes.

In the Chinese study of one hundred and seventy-two children, the leading genes were ATP seven B, which causes Wilson disease, SLC twenty-five A thirteen and G six PC, which causes glycogen storage disease type one. In children with an enlarged liver and spleen, the yield was eighty-four point six percent.

In the Turkish study, where many parents are related to each other, the genes were mostly recessive: ATP eight B one, ABCB eleven, ABCB four, DCDC two, DGUOK, KIF twelve, USP fifty-three and HSD three B seven. Many of these cause low-GGT cholestasis, which we will come back to later.

And in the European study of acute liver failure, the leading genes were NBAS, MPV seventeen and DGUOK. Forty-five percent of the diagnoses were mitochondrial disease, and twenty-eight percent were disorders of protein transport inside the cell. The lesson is that acute liver failure needs an exome or a genome, not a cholestasis panel, because most of these genes are not on a cholestasis panel.

Part three. What a negative result means.

A negative genetic result means negative for what was looked at, by the method that was used. There are four common reasons why a real genetic disease can hide behind a negative panel.

The first reason is copy number. Of about seven hundred known JAG one variants, about seventy are large deletions. Sequencing alone can miss them. In Alagille syndrome, about three point two percent of children with a clear clinical diagnosis still have no variant found. So when a panel comes back negative, always ask whether deletion and duplication analysis was included.

The second reason is newer genes. Genes such as KIF twelve, USP fifty-three and ZFYVE nineteen were linked to cholestasis after many panels were already built. In the Japanese study by Ito, an expanded panel solved five point two percent of children whose earlier, older panel had been negative.

The third reason is non-coding variants. Variants deep inside an intron, or in the region that switches a gene on, are not captured by an exome. When the clinical picture is convincing and the exome is negative, a genome or an RNA study may find the answer.

The fourth reason is phase. A child may have two variants in a recessive gene, but if both sit on the same copy, the child is only a carrier. Only testing the parents can tell you. So parental testing can turn two variants into a diagnosis, or into a carrier state.

Part four. Which test to send, and why to send the parents.

In 2018, Clark and colleagues pooled thirty-seven studies with twenty thousand and sixty-eight children with suspected genetic disease. They compared the three main

platforms. The chromosomal microarray found a diagnosis in ten percent of children and changed management in six percent. The exome found a diagnosis in thirty-six percent and changed management in seventeen percent. The genome found a diagnosis in forty-one percent and changed management in twenty-seven percent.

The exome and the genome were not significantly different from each other. But both were much better than the microarray, and the genome changed management more often than the microarray.

The same analysis had one more interesting finding. When the results were interpreted by a team in the hospital, the yield was forty-two percent. When they were interpreted by a distant reference laboratory, it was twenty-nine percent. The difference is the clinical information. The laboratory can only interpret variants against the picture of the child it is given. So always send a good clinical description with the sample.

Now to the trio. In the same analysis by Clark and colleagues, when studies compared sequencing the child with the parents against sequencing the child alone, the odds of a diagnosis were about two times higher with the trio. The odds ratio was two point zero four. That is why people say that the parents are the cheapest functional test in genetics.

The trio helps in three ways. First, it shows the phase. It proves that two recessive variants sit on different copies of the gene. Second, it shows when a variant is de novo. A new dominant variant in JAG one or NOTCH two, absent in both healthy parents, can be classified as disease-causing much faster. Third, it reduces uncertainty. If a candidate dominant variant is also present in a healthy parent, it is less likely to be the cause.

The EASL guideline of 2024 says that testing affected and unaffected relatives is one way to upgrade a variant of uncertain significance to likely pathogenic. Here is a practical rule. If the child alone was sequenced and two variants in ABCB eleven were found, that is a hypothesis, not a diagnosis, until the parents have been tested.

Part five. The sick infant and rapid genome sequencing.

In a sick child, speed is part of the value of the test. A result that arrives after the decision has been made is of little use.

Project Baby Bear, reported by Dimmock and colleagues in 2021, offered rapid whole-genome sequencing to infants under one year old in five intensive care units in California. Seventy-four of one hundred and eighty-four infants, or forty percent, received a diagnosis. The median time to diagnosis was three days. Medical care changed in thirty-two percent of the infants. The study spent about one point seven million dollars on testing, and the estimated saving in hospital costs was between two point two and two point nine million dollars.

A smaller study by Bonser and colleagues, in 2023, looked at children with acute liver dysfunction. Seven of eighteen children, or thirty-nine percent, received a diagnosis, with a median of eight days. Results came back faster with the rapid genome, in about four days compared with ten. Interestingly, four children with a negative result turned out to have a toxic exposure. So a negative rapid genome should make you look again for a drug or a toxin.

In England, the NHS test directory makes this routine. Test R fourteen is a rapid trio genome for very sick children in neonatal and paediatric intensive care, with a preliminary result within two weeks.

Part six. Acute liver failure of unknown cause.

In paediatric acute liver failure, a large group of children have no cause found. This is called indeterminate acute liver failure. The study by Lenz and colleagues, published in 2024, suggests that indeterminate often means not yet sequenced.

They performed exome sequencing in two hundred and sixty children with indeterminate acute liver failure from nineteen countries. Thirty-seven percent received a genetic diagnosis. When the liver failure started in the first year of life, the yield was forty-one percent. And in fifty-nine children with recurrent episodes of acute liver failure, the yield was sixty-four percent.

Forty-five percent of the diagnoses were mitochondrial. The most common genes were NBAS, in twenty children, MPV seventeen, in eight, and DGUOK, in seven.

The outcomes were serious. About one third of the children died, and fifty-six had a liver transplant. In many of them, the genetic result arrived after the transplant decision it should have informed. The authors concluded that every child with indeterminate acute liver failure should have exome sequencing now, and in future a rapid genome.

The practical rule is this. In indeterminate acute liver failure, especially in the first year of life or when it recurs, send sequencing before or alongside listing for transplant. But if the child already meets transplant criteria, do not wait for the result.

Part seven. Reading the result: the five classes of variant.

The ACMG standards of 2015, written by Richards and colleagues, sort every variant into one of five classes. It helps to think of them as five tiers that lead to three actions.

A pathogenic variant causes disease, and there is enough evidence to act. You make the diagnosis, treat, and offer testing to relatives. This testing of relatives is called cascade testing.

A likely pathogenic variant is, by convention, at least ninety percent certain to cause disease. You act as if the diagnosis is made, provided the clinical picture fits.

A variant of uncertain significance, usually shortened to VUS, is neither proven harmful nor proven harmless. Most of these are missense variants, which change one amino acid in the protein. You do not make a diagnosis on a VUS. Instead, you work to resolve it, and you plan to recontact the family when it is reclassified.

A likely benign variant is at least ninety percent certain to have no effect. You do not act on it, and it is usually not reported.

A benign variant has population or functional proof that it is harmless. You ignore it.

In real life, the commonest dilemma is a recessive gene in which the child has one pathogenic variant and one VUS. In the study by Hoskins and colleagues, this situation, reported as a potential diagnosis, occurred in two point two percent of ten thousand eight hundred and ninety-four panels. The genes most often involved were ABCC two, ABCB four, ABCB eleven, CFTR and PKHD one.

Part eight. The burden of uncertainty.

The more genes you test, the more uncertain variants you find. In the American study by Karpen and colleagues, two thousand one hundred and seventy-one cholestasis panels produced three thousand one hundred and seventeen VUS, compared with only six hundred and sixty-two pathogenic or likely pathogenic variants. That is nearly five uncertain variants for every clear one.

In a larger study by Chen and colleagues in 2023, forty-one percent of one point six nine million people who had genetic testing carried at least one VUS. Of the VUS that were later reclassified, eighty percent turned out to be benign or likely benign. It took on average twenty-two months for a VUS to be upgraded to pathogenic, and thirty point seven months to be downgraded to benign.

There is also an important problem of fairness. The number of VUS per gene was higher in every group that was not of European white ancestry. This is because the reference databases contain mostly people of European ancestry. So a child from an under-represented background is more likely to receive an uncertain result.

An earlier study by Mersch and colleagues, in 2018, found that seven point seven percent of unique VUS were reclassified over time, and ninety-one percent of those were downgraded. The message is simple. Most VUS are benign variants that are still waiting for proof.

Part nine. How to resolve a VUS.

The good news for hepatologists is that the liver has its own tests. The phenotype is evidence. Send it to the laboratory together with the sample. There are four ways to gather more evidence.

The first is the family. Test the parents to establish the phase. Look at whether the variant follows the disease in affected and healthy siblings. And find out whether the same variant has been seen in other affected families.

The second is the tissue. On a liver biopsy, immunostaining can show whether the BSEP protein, made by ABCB eleven, or the MDR three protein, made by ABCB four, is present or missing. The EASL guideline notes that a biopsy can help in suspected Alagille syndrome with a VUS. And the GGT and serum bile acids should fit the gene. For example, BSEP deficiency causes cholestasis with a low GGT.

The third is biochemistry. Urine bile acids identify bile acid synthesis defects. Succinylacetone identifies tyrosinaemia type one. Citrulline points to citrin deficiency. Copper studies support Wilson disease. And the phospholipid level in bile supports ABCB four disease.

The fourth is RNA and function. A splice minigene assay or RNA sequencing can show whether a variant disrupts splicing. In vitro transport assays can measure whether the protein works. Computer prediction tools, called in silico tools, give only supporting evidence. They are never enough on their own.

Part ten. Reanalysis: a negative exome is a draft.

An exome or genome can be read again. A panel cannot, because it never read the other genes. This is the main practical reason to prefer exome or genome when the diagnosis is unclear.

In a meta-analysis of twenty-nine studies, Dai and colleagues, in 2022, found that reanalysing old exome data gave a further diagnosis in about ten percent of previously negative cases. The yield was higher when more than twenty-four months had passed. Most new diagnoses came from updates in the medical literature, such as the discovery of a new disease gene. For example, in 2024 Zheng and colleagues re-read old exomes and found KIF twelve and USP fifty-three cholestasis once those genes had been published.

The EASL guideline of 2024 recommends that exome or genome data be reanalysed at least every three years. The ACMG statement of 2019 says that laboratories must have a reanalysis policy. But it does not set an interval, and it does not say who should ask for the reanalysis. In practice, that person is the clinician.

Reclassification also works in the other direction. Kobayashi and colleagues, in 2024, followed two point zero five million variants over eight years. Four point six percent were reclassified, but only zero point zero six percent were reversed from one side to the other, for example from pathogenic to benign. So a clear result is very rarely overturned.

Part eleven. When the gene changes the plan.

Genetic testing is worth doing when the result changes what you do. The best example in paediatric hepatology is BSEP deficiency, caused by biallelic variants in ABCB11. This disease used to be called progressive familial intrahepatic cholestasis type two.

The NAPPED consortium, reported by van Wessel and colleagues in 2020, studied two hundred and sixty-four children with severe BSEP deficiency. They divided the children into three groups by genotype.

BSEP one includes children with at least one of two specific milder missense variants, p dot E two nine seven G or p dot D four eight two G. In these children, the median survival with their own liver was twenty point four years, and the risk of liver cancer by age fifteen was four percent.

BSEP two includes children with other missense variants. Their median survival with their own liver was seven years, and the risk of liver cancer by age fifteen was seven percent.

BSEP three includes children with two truncating variants, which means the protein is cut short and is essentially absent. Their median survival with their own liver was only three point five years, and the risk of liver cancer by age fifteen was thirty-four percent.

Surgical biliary diversion helped only the BSEP one and BSEP two groups, with a hazard ratio of zero point five zero, which means the risk of losing the native liver was halved. If the serum bile acids fell below one hundred and two micromoles per litre after diversion, the child was likely to keep their own liver for at least fifteen years.

The EASL guideline of 2024 makes a strong recommendation, based on level three evidence, that follow-up, screening for liver cancer, and the decision about biliary diversion should be based on the specific genotype.

This links to the newer drugs, the ileal bile acid transporter inhibitors, often shortened to IBAT inhibitors. They block the reabsorption of bile acids in the gut. There are two: maralixibat, sold as Livmarli, and odeixibat, sold as Bylvy. In BSEP three, where the protein is absent, these drugs are not expected to work. The American regulator, the FDA, lists this as a limitation of use, and the European regulator, the EMA, includes it as a warning. In BSEP one, on the other hand, both diversion and an IBAT inhibitor are worth trying, and transplant may be decades away.

There are other examples where the gene changes the plan.

NBAS deficiency causes episodes of acute liver failure triggered by fever. In a study of two hundred and thirty patients by Peters and colleagues, in 2025, the crises became less frequent with age, and survival with the native liver was eighty-three point nine percent. So the right first step is an emergency protocol, with aggressive fever control and early glucose, not an early transplant.

In a case report from 2017, rapid trio exome sequencing found Niemann–Pick type C, caused by NPC one, in an infant with neonatal liver failure. The result took the infant off the transplant list.

POLG disease, a mitochondrial disorder, matters in two ways. Valproate is contraindicated, because it can trigger fatal liver failure, and this is a boxed warning from the FDA. And the neurological course of the disease weighs heavily on whether transplant is appropriate.

Finally, TJP two disease and ABCB eleven disease both carry a risk of liver cancer. The genotype therefore decides how often you image the liver and measure the alpha-fetoprotein.

Part twelve. Do not wait for the gene.

This is perhaps the most important part of the chapter for a first-year fellow. For several inherited liver diseases, there is a fast clue and an effective treatment. In these cases you treat first and let the gene confirm later.

Biliary atresia is not a single-gene disease, but it is the first thing to exclude in any infant with cholestasis. The clues come within hours: pale or acholic stools, a raised GGT, the ultrasound, and the serum MMP seven. Proceed straight to the cholangiogram and the Kasai pathway. A cholestasis panel can be sent at the same time, in parallel. Never delay the cholangiogram to wait for a panel. Sometimes a JAG one result later reframes a poor outcome after Kasai, because the child actually had Alagille syndrome.

In galactosaemia, the clues are a low GALT enzyme activity, which must be measured before any blood transfusion, and sepsis with *E. coli*. Stop lactose as soon as you suspect it.

In tyrosinaemia type one, the clue is succinylacetone in the urine or blood. Start nitisinone.

In bile acid synthesis defects, the clues are cholestasis with a low GGT, normal or low serum bile acids, and abnormal urine bile acids. Start oral cholic acid.

Gestational alloimmune liver disease is not a genetic disease at all, but it can look like one. The clues are a very high ferritin, iron deposits outside the liver seen on MRI or on a

lip biopsy, and staining for the complement complex C five b nine in the liver. Treat with intravenous immunoglobulin, with or without exchange transfusion.

In citrin deficiency, the clues are a high citrulline, raised galactose, and a fatty liver. Change to a lactose-free formula rich in medium-chain triglycerides.

And in recurrent acute liver failure with fever, the pattern alone suggests NBAS. Treat the fever, give glucose early, and follow an NBAS emergency protocol while the sequencing is done in parallel.

Part thirteen. What the guidelines say.

Let us bring the guidelines together. On most points they agree, but the European and British documents go further than the North American ones.

On when to test. The NASPGHAN and ESPGHAN guideline of 2017 on the jaundiced infant advises a low threshold for a panel or an exome, in collaboration with the hepatologist. This advice is not graded. The EASL guideline of 2024 goes further. It recommends genetic testing early in the work-up of infants and children, and after the common causes have been excluded in adults.

On which test. The ACMG statement of 2021 recommends exome or genome sequencing as a first- or second-tier test, but for children with congenital anomalies or developmental delay and intellectual disability, not for liver disease. So applying it to liver disease is an extrapolation. The EASL guideline recommends a next-generation sequencing panel or an exome, discourages Sanger sequencing gene by gene, and suggests a genome if the exome is negative.

On who qualifies. In North America there are no national criteria, and testing is mostly decided by what insurers will pay for. In England, test R one seven one covers children under eighteen with unexplained cholestasis lasting more than three months or recurring, after review by a national multidisciplinary team.

On reanalysis. The ACMG statement of 2019 requires each laboratory to have a policy, but sets no interval and does not say who should ask. EASL recommends reanalysis at least every three years.

On counselling, the guidelines agree. Both ACMG and EASL say that counselling must come before testing, and that the policy on incidental or secondary findings must be agreed with the family in advance.

And on BSEP three and the IBAT inhibitors, the drug labels differ. The FDA says the drugs are not recommended in BSEP three, as a limitation of use. The EMA says these patients are not expected to respond, as a warning. The licensed ages for progressive familial intrahepatic cholestasis also differ. In the United States, maralixibat is licensed

from twelve months and odevixibat from three months. In Europe, maralixibat is licensed from three months and odevixibat from six months.

Putting it into practice.

Let us finish with eight common situations.

An infant with pale, acholic stools. Follow the biliary atresia pathway first, and send a panel in parallel. A JAG one result may later explain a poor result after Kasai. Do not delay the cholangiogram for a panel.

An infant with cholestasis in whom biliary atresia has been excluded. Send a broad panel that includes deletion and duplication analysis, or a trio exome, together with urine bile acids. A low GGT points to ATP eight B one, ABCB eleven or a bile acid synthesis defect. Do not order genes one at a time.

A child with syndromic features, such as heart disease, butterfly vertebrae or a typical face. Test JAG one and NOTCH two, including deletion analysis. A negative sequence is not the same as a negative test for Alagille syndrome. Do not stop at a negative sequence.

A child with indeterminate or recurrent acute liver failure. Send a rapid trio genome, for example through test R fourteen. A mitochondrial or NBAS result changes transplant candidacy. But do not wait for the result if the child already meets transplant criteria.

An older child with an unexplained raised ALT or cholestasis. Send an exome with a liver-focused analysis. Think of ATP seven B, ABCB four, and heterozygous variants that act as co-factors. Do not call a single ABCB four variant the cause of the disease.

A child with one pathogenic variant and one VUS in a recessive gene. Test the parents, consider immunostaining of the biopsy, and use biochemistry. Treat this as a hypothesis, not a diagnosis. Do not offer cascade testing to relatives on the basis of the VUS.

A child with confirmed BSEP deficiency. Classify the genotype as BSEP one, two or three. This guides the use of IBAT inhibitors, biliary diversion and liver cancer screening. Do not start an IBAT inhibitor in BSEP three.

And a child with a negative exome whose disease persists. Ask for reanalysis within three years, or move to a genome. A negative exome is a draft, not a verdict. Do not close the file.

Key points.

The yield of genetic testing reflects who was tested. It was seventy-one percent in children whose disease looked genetic, about seven percent in open referral, and four percent when prematurity explained the picture.

A broad panel and an exome perform about equally. But an exome or a genome can be read again later, and a panel cannot.

Send the trio. The odds of a diagnosis double, and only the parents can show whether two recessive variants sit on different copies of the gene.

Indeterminate acute liver failure in children is genetic in about one third, and in about two thirds when it recurs. Sequence before or alongside listing for transplant.

A VUS is not a diagnosis. Most VUS that are resolved turn out to be benign, and it takes about two years.

One pathogenic variant plus one VUS in a recessive gene is the commonest dilemma. Phase it with the parents, then test the tissue and the biochemistry.

Reanalyse a negative exome at least every three years, as EASL recommends. Expect about one new diagnosis in every ten.

Never let a pending genetic result delay the cholangiogram, nitrisinone, lactose withdrawal, cholic acid or intravenous immunoglobulin.

Questions to test yourself.

Question one. Why did the definitive yield of the American cholestasis panel fall from twelve percent to six point eight percent? Answer: because the test was opened to all comers. The yield depends mainly on how carefully the children were selected, not on the quality of the test.

Question two. A panel for Alagille syndrome is negative in a child with typical clinical features. What should you ask the laboratory? Answer: whether deletion and duplication analysis was included. About seventy of roughly seven hundred known JAG one variants are large deletions that sequencing alone can miss.

Question three. Give three reasons to send the parents' samples together with the child's. Answer: to establish the phase of two recessive variants, to show that a dominant variant is de novo, and to downgrade a candidate variant found in a healthy parent. Overall, a trio roughly doubles the odds of a diagnosis, with an odds ratio of two point zero four.

Question four. In the European study of indeterminate acute liver failure by Lenz and colleagues, what was the genetic yield overall, in the first year of life, and in recurrent cases? Answer: thirty-seven percent overall, forty-one percent when the illness began in the first year of life, and sixty-four percent in recurrent acute liver failure. Forty-five percent of the diagnoses were mitochondrial.

Question five. What proportion of reclassified variants of uncertain significance turn out to be benign, and how long does an upgrade to pathogenic take on average? Answer:

about eighty percent turn out benign or likely benign. An upgrade to pathogenic takes on average about twenty-two months.

Question six. Name four ways to gather evidence about a VUS in a cholestasis gene.

Answer: family studies, such as parental testing and segregation in siblings; tissue, such as BSEP or MDR three immunostaining on a biopsy; biochemistry, such as urine bile acids, GGT or biliary phospholipids; and RNA or functional assays. Computer prediction tools give only supporting evidence.

Question seven. How often does EASL recommend reanalysis of a negative exome, and what extra yield can you expect? Answer: at least every three years. A meta-analysis of twenty-nine studies found about ten percent extra yield, higher after twenty-four months.

Question eight. In BSEP deficiency, what are the median native liver survival and the risk of liver cancer by age fifteen in the BSEP three group, and which treatment should you avoid? Answer: a median native liver survival of three point five years and a thirty-four percent risk of liver cancer by age fifteen. Biliary diversion did not prolong native liver survival, and an IBAT inhibitor should not be started, because no response is expected.

Question nine. Name four inherited or inherited-looking liver diseases in which you should treat before the genetic result arrives, and the treatment for each. Answer: galactosaemia, stop lactose; tyrosinaemia type one, start nitisinone; bile acid synthesis defects, start oral cholic acid; and citrin deficiency, change to a lactose-free formula rich in medium-chain triglycerides. Gestational alloimmune liver disease is not genetic, but it also needs immediate treatment with intravenous immunoglobulin.

End of chapter.